

Explicit Semantic Modeling for Solving Word Association Puzzles: A Multi-Signal Ensemble Approach

Aayush Pandey
2025201058

Srushti Pekamwar
2025201066

Suparshwa Patil
2025201090

Anurag Kacholiya
2025202025

Abstract

Word connection puzzles, such as the New York Times *Connections*, require partitioning 16 words into four disjoint groups based on latent semantic, contextual, or structural relations. Standard NLP models whether embedding-based clustering or large language models fail systematically on this task due to deliberate “red herring” distractors, polysemy, and the combinatorial nature of global constraint satisfaction. This paper presents our final solution: a **Multi-Signal Similarity Ensemble** that fuses 17 heterogeneous linguistic signals, including contextual embeddings contrastively fine-tuned via triplet loss, knowledge-graph-augmented representations (ConceptNet Numberbatch), phonetic similarity, and grid-aware uniqueness matrices. An ML ensemble of Random Forest, Gradient Boosting, and LightGBM LambdaRank classifiers scores candidate groups, with a global joint optimizer selecting the partition that maximizes total ensemble agreement. Trained on a data-augmented corpus of 1,278 puzzles (original + LLM-generated + BabelNet-sourced), our final model achieves **33% full-game accuracy** and **50.25% group accuracy** on the held-out 100-puzzle test set a $2\times$ improvement over our Ensemble-Itr-1 baseline demonstrating that heterogeneous signal fusion combined with global optimization is the decisive factor in closing the gap with human-level lateral reasoning.

1 Introduction

Word association and grouping tasks are a cornerstone of human language understanding and abstract reasoning. The New York Times *Connections* puzzle (see Figure 1) presents a deceptively challenging problem: given a board of exactly 16 words, a solver must partition them into four groups of exactly four words each, where each group shares a hidden thematic, semantic, lexical, or structural relationship. Groups are designed to be mutually misleading words in one group often appear to belong to another, testing not just semantic knowledge, but lateral thinking and relational constraint satisfaction.

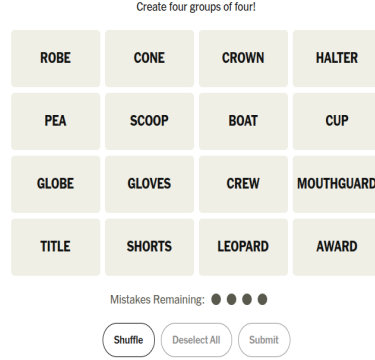


Figure 1: An example NYT *Connections* puzzle

1.1 Motivation

Current NLP systems fail on this task for three compounding reasons. First, **distributional embeddings** are vulnerable to polysemy: a single word may be semantically close to members of two different groups simultaneously. Second, **greedy clustering** is irrecoverable: once a wrong word enters a group, the cascade error contaminates all subsequent guesses. Third, **LLM-only approaches** are opaque they cannot articulate the precise relational constraint that defines each group, making them unreliable on adversarial lateral-thinking categories. Figure 2 illustrates how the correct color-coded groupings expose hidden relational structure that surface-level similarity cannot recover.

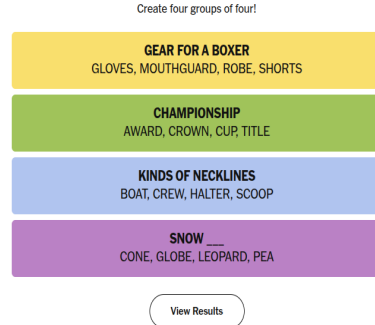


Figure 2: The solved state of the puzzle from Figure 1.

Our investigation begins with the central hypothesis that *no single linguistic signal is sufficient*; rather, solv-

ing *Connections* requires the simultaneous interrogation of semantic, lexical, phonetic, ontological, and encyclopedic knowledge, fused under a combinatorial global optimizer that enforces exact disjoint partitioning.

1.2 Objectives

This project is structured into two progressive components:

- **Part I:** Establish strong empirical baselines using embedding-based clustering and Graph Neural Networks (GNNs) to quantify the task’s difficulty and identify primary failure modes.
- **Part II (Core Contribution):** Develop a deterministic, multi-signal similarity ensemble pipeline capturing diverse linguistic constraints (semantic, lexical, phonetic, ontological) and solve puzzles systematically using combinatorial global optimization, data augmentation, and contrastive fine-tuning.

2 Literature Review

The *New York Times Connections* game has evolved from a popular human puzzle into a rigorous benchmark for computational reasoning, motivating exploration of explicit semantic modeling.

2.1 Benchmarking Lateral Reasoning and Abstraction

Initial studies established *Connections* as a stress test for LLMs. **NYT-Connections** (2) introduced the game as a classification benchmark, demonstrating that even GPT-4 falls significantly short of human performance under isolated System-1 inference. **Connecting the Dots** (6) analyzed over 400 puzzles, proving that encyclopedic, multiword, and form-based structural knowledge are all necessary for success. **Missed Connections** (10) framed the task as a direct measure of lateral thinking, demonstrating that embedding-based clustering and zero-shot LLMs consistently fail without deeper relational constraint satisfaction.

2.2 Knowledge-Augmented Semantic Representations

Purely distributional embeddings are vulnerable to semantic distractors. **MPNet** (8) addresses limitations of prior contextual models by unifying Masked and Permuted Language Modeling, yielding highly robust sentence-level embeddings. **ConceptNet 5.5** (9) introduced Numberbatch, which retrofits structured knowledge-graph constraints into distributional vectors, significantly outperforming text-only models on semantic relatedness. For our task, we concatenate fine-tuned MPNet embeddings (768-dim) with Numberbatch vectors (300-dim) into a 1,068-dimensional joint representation that merges task-optimized contextual semantics with explicit world knowledge.

2.3 Word Sense Disambiguation and Contextual Embeddings

A critical failure mode for word association puzzles is polysemy. **WordNet** (5) provides a rich lexical database with synset hierarchies, enabling Wu-Palmer similarity computations that respect fine-grained semantic distinctions. To further handle polysemy at inference time, we implement a multi-sense embedding strategy: for each word, we generate a weighted average over all its WordNet synsets, encoding the word across its entire sense distribution rather than anchoring to a single representation an approach empirically supported by recent work on sense-aware embeddings (4).

2.4 Graph Neural Networks in Combinatorial Optimization (Brief Overview)

Early in this project, we explored R-GCNs for the grouping task. While GNNs theoretically capture multi-hop lexical relationships, **Optimizing over trained GNNs via symmetry breaking** (11) highlights that extracting globally valid disjoint partitions from continuous GNN outputs requires mixed-integer programming a prohibitive computational overhead for our domain. Our empirical results confirmed this: GNN-based models achieved 0% full-game accuracy due to graph sparsity and arbitrary label assignments, motivating our pivot to the ensemble approach.

3 Methodology

Our final approach is a **Multi-Signal Similarity Ensemble** with three principal stages: (1) rich multi-modal word representation, (2) ML ensemble scoring, and (3) global joint optimization via beam search. The full architecture is illustrated in Figure 3.

3.1 System Architecture

The system is organized into four conceptual stages that operate in a strict sequence. First, a **multi-modal embedding module** encodes each of the 16 puzzle words using complementary linguistic resources contextual language models, static word vectors, knowledge-graph embeddings, and phonetic dictionaries. Second, a **pair-wise similarity computation stage** constructs 17 independent 16×16 similarity matrices, each capturing a distinct relational signal between word pairs. Third, a **feature extraction and ML scoring stage** converts each candidate 4-word group into a rich feature vector and scores it using a trained ensemble of classifiers and a learning-to-rank model. Finally, a **global joint optimizer** selects the board partition that maximizes the combined ensemble score across all four groups simultaneously, enforcing the strict game constraint of exactly four disjoint groups of four.

3.2 Multi-Modal Word Representation

Each word on the 16-word board is encoded through multiple complementary lenses to ensure that no single linguistic perspective dominates:

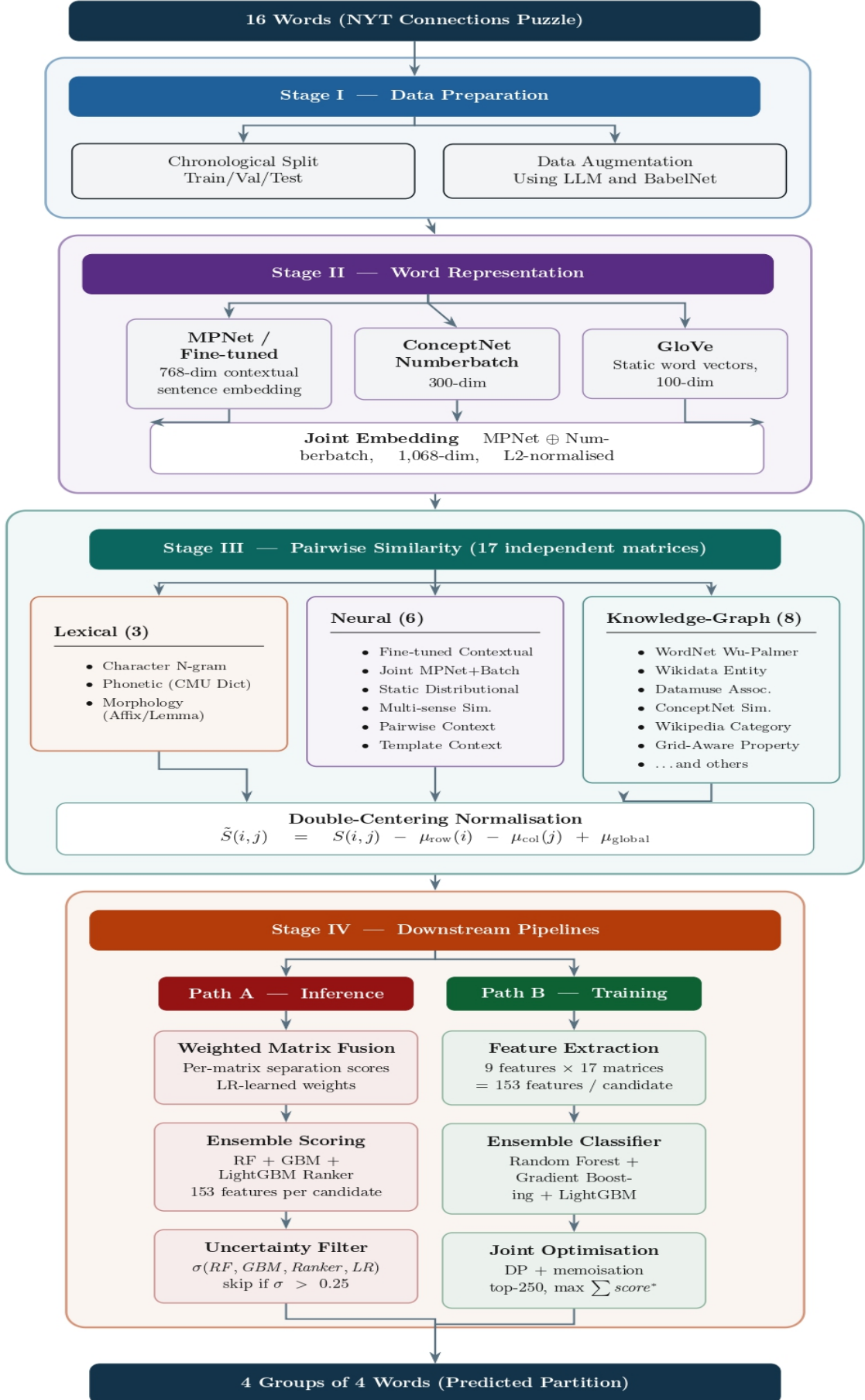


Figure 3: Architecture diagram of the proposed Multi-Signal Similarity Ensemble.

- **Contrastively Fine-Tuned Contextual Semantics:** A pre-trained sentence transformer (MPNet-768) is fine-tuned on the training set using triplet loss with hard-negative mining. For each anchor word, the hardest negative is the most similar out-of-group word under the current model. This adaptation pulls true group members closer together in the embedding space while pushing distractors apart, directly optimizing for the grouping task rather than general semantic similarity.
- **Static Distributional Semantics:** GloVe word vectors (100-dim) capture broad co-occurrence statistics that are complementary to contextual embeddings, providing a stable, task-agnostic semantic baseline.
- **Knowledge-Graph Augmented Embeddings:** ConceptNet Numberbatch (300-dim) retrofits structured common-sense knowledge into distributional vectors. The contextual and Numberbatch embeddings are concatenated and re-normalized to unit norm, yielding a joint representation that merges task-optimized semantics with explicit world knowledge.
- **Multi-Sense Embeddings:** Each word is separately encoded for every WordNet synset it belongs to, and the representations are aggregated with sense-frequency weighting. This allows polysemous words (e.g., *Mercury* as a planet, element, car brand, or mythological figure) to carry signal for multiple possible group memberships simultaneously.

3.3 Pairwise Similarity Matrix Construction

For each 16-word board, 17 independent 16×16 similarity matrices are constructed, each capturing a distinct relational signal. The matrices are grouped into three conceptual families:

3.3.1 Lexical Similarity

Lexical signals measure the surface form of words, independent of their meaning, making them crucial for puzzle categories based on orthographic or phonetic patterns:

- **Character N-gram Similarity**
- **Phonetic Similarity**
- **Morphological Similarity**

3.3.2 Neural Embedding Similarities

Neural signals measure distributional and contextual meaning, capturing thematic and conceptual relatedness that surface forms cannot express:

- **Fine-tuned Contextual Similarity**
- **Joint Contextual - MPNet + NumberBatch Similarity**
- **Static Distributional Similarity**

- **Multi-Sense Similarity**
- **Pairwise Context Similarity**
- **Template-Context Similarity**

3.3.3 Knowledge-Graph Based Similarities

Knowledge-graph signals leverage external structured resources—ontologies, encyclopedias, and association databases—to capture explicit real-world relationships:

- **WordNet Similarity**
- **Wikidata Entity Similarity**
- **Datamuse Association Similarity**
- **ConceptNet Similarity**
- **Wikipedia Category Similarity**
- **Grid-Aware Property Uniqueness**
- **Grid-Aware Category Uniqueness**
- **Grid-Aware Association Uniqueness**

To eliminate **hubness**—the systematic bias whereby high-frequency words appear artificially similar to all others—each matrix is independently *Double-Centered* before fusion:

$$\tilde{S}(i, j) = S(i, j) - \mu_{\text{row}}(i) - \mu_{\text{col}}(j) + \mu_{\text{global}} \quad (1)$$

3.4 Feature Engineering

For each candidate 4-word group S , we extract 9 features per matrix (153 total features at full configuration):

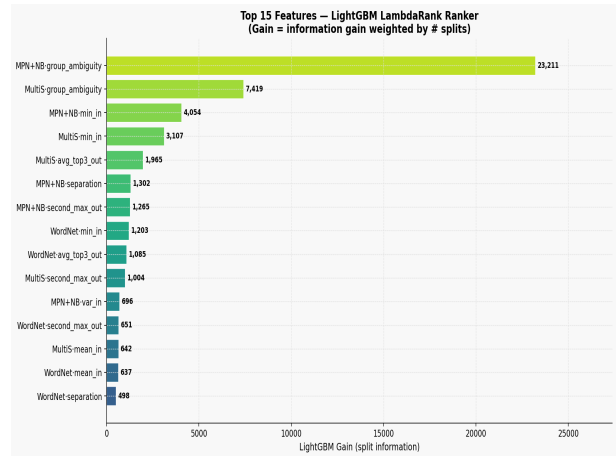


Figure 4: Top 15 Features Extracted (out of 153)

- **In-group cohesion** (mean_in, min_in, var_in): Measures how tightly the 4 candidate words cluster together across all $\binom{4}{2} = 6$ intra-group edges. For example, {Bass, Trout, Salmon, Carp} should yield a high mean_in on WordNet_sim since all are fish hyponyms.
- **Distractor separation** (max_out, second_max_out, avg_top3_out, separation = mean_in - mean_out): Captures how well the candidate group stands apart from all other 12 words on the board. A high max_out signals a dangerous “red herring” a word outside the group that is deceptively close to its members.

- **Group ambiguity:** The sum of each candidate word’s average similarity to every out-of-group word. High ambiguity flags groups where members “fit” multiple plausible clusters a classic failure mode for polysemous words like *Mercury* (planet, element, car, myth).
- **Triangle consistency:** Fraction of consistent triplets within the group if $A \sim B$ and $B \sim C$, transitivity demands $A \sim C$. Violated transitivity (e.g., *Spring* links to both *Summer* and *Coil* but *Summer* and *Coil* are unrelated) strongly indicates a false grouping.

3.5 ML Ensemble Scoring

Here is the Full 3-stage Pipeline :

Stage 1 Candidate Generation: All $\binom{16}{4} = 1,820$ possible 4-word subsets are enumerated as candidates.

Stage 2 ML Ensemble Scoring:

- **Random Forest** (200 trees, max_depth=10): A Gini-gain ensemble classifier trained on positive groups and a mix of hard (3-overlap) and easy random negatives.
- **Gradient Boosting** (200 estimators, max_depth=3): A complementary sequential weak-learner ensemble resistant to overfitting, with gradient-based loss minimization.
- **LightGBM LambdaRank** (500 boosting rounds, lambdarank objective, NDCG@{1,3,5}): A pairwise learning-to-rank model that learns which candidate group is more likely to be correct relative to all others within the same puzzle a direct formulation of the ranking problem. **NDCG (Normalized Discounted Cumulative Gain)** measures ranking quality by rewarding models that promote correct groups toward the top of the ranked list, discounting gains logarithmically by position; an NDCG@1 of 0.92 means the ranker identifies the single best group correctly for over 90% of validation puzzles.
- **Logistic Regression Matrix Fusion:** Trained separately on the validation set to learn a weighted combination of per-matrix separation scores, producing an interpretable per-signal importance vector that reveals which similarity signals contribute most to puzzle solving.

Final ensemble combination: $0.15 \cdot p_{RF} + 0.15 \cdot p_{GBM} + 0.50 \cdot p_{\text{LambdaRank}} + 0.20 \cdot p_{\text{PLR}}$.

Stage 2b Competition-Aware Adjustments: Adjusted scores incorporate a rank bonus (α/rank) and a gap bonus ($\beta \cdot \text{gap_to_next}$), preventing a single dominant high-scoring group from crowding out equally valid candidates.

Stage 3 Global Joint Optimization: Rather than greedy selection, the solver jointly finds the 4-partition of 16 words that maximizes the sum of adjusted scores across all four groups simultaneously. This prevents the

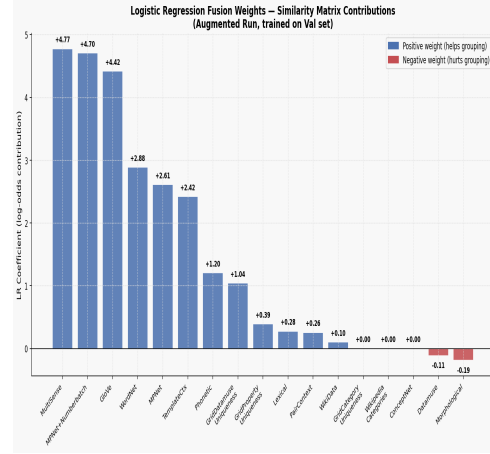


Figure 5: Logistic Regression Fusion Weights.

“blocking” failure mode where one early commitment eliminates valid completions for the remaining words. An uncertainty filter withholds greedy commits when model agreement is low, defaulting to the global joint solve.

3.6 Data Augmentation

With only 678 training puzzles in the original dataset (yielding ~2,712 positive group examples), there is a severe risk of overfitting for an ML ensemble with 153 features. Two complementary augmentation strategies expand the training corpus:

3.6.1 Method 1: LLM-Based Puzzle Generation

An automated two-phase pipeline generates structurally valid novel puzzles (see Figure 6 for a sample). Phase 1 encodes the game mechanics, lateral wordplay patterns, and group-exclusivity rules into a structured few-shot **Meta-Prompt**. Phase 2 uses a generative LLM to produce candidate puzzles, with automated validation enforcing structural correctness (exactly four disjoint groups of four unique words, no lexical overlap). This process yields 100 additional structurally valid training puzzles.



Figure 6: A sample puzzle generated by Method 1

3.6.2 Method 2: BabelNet Knowledge-Graph Augmentation

A deterministic pipeline synthesizes 500 taxonomically precise puzzles by querying BabelNet (sense disambiguation), WordNet (hyponym/meronym trees), Datamuse (thematic associations), and the Brown Corpus (frequency filtering) (see Figure 7 for a sample). Four difficulty tiers (L1–L4) are generated with Jaccard synset filtering for semantic exclusivity, and an L4 compound-stem tier deliberately introduces controlled “red herrings.”

Final training set: 678 (original) + 500 (BabelNet) + 100 (LLM) = **1,278 puzzles**. Val and test sets remain original-only (137 and 100 games respectively) to ensure uncontaminated evaluation.

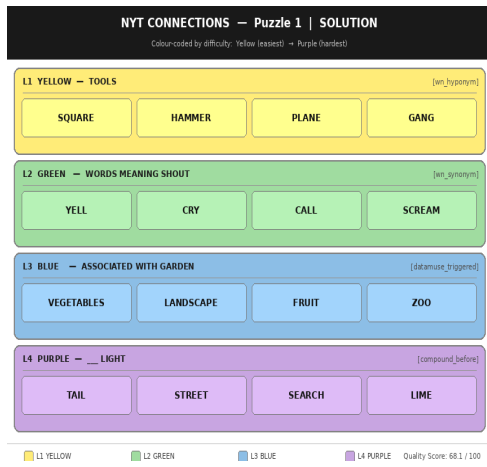


Figure 7: A sample puzzle generated by Method 2

4 Evaluation Strategies

Models are evaluated against the held-out 100-puzzle test set using strict combinatorial metrics:

1. **Game Accuracy:** Fraction of full games solved completely (all 4 groups correctly identified in at most 4 guesses).
2. **Grouping Accuracy:** Total individual groups correctly identified out of 400 possible across the test set.
3. **One-Away Metric:** Number of groups where exactly 3 of 4 words were correctly clustered indicating the model identified the correct latent relationship but failed at final single-word disambiguation.

It is important to note that our system operates under a strict **4-guess budget**, which is the actual game constraint. Some prior work in the literature (e.g., (10)) evaluates models with a relaxed 5-guess budget, which inflates success rates. All our results are reported strictly at the 4-guess limit.

5 Timeline

- **January 2026:** Project initiation, dataset acquisition (914 NYT Connections games, chronological split). Implementation and evaluation of Baselines 0–2 (GloVe+K-Means, GloVe+SBERT agglomerative clustering, MPNet with Double Centering).
- **February 2026:** Graph Neural Network exploration (Relational-GCN node classification and global pairwise scoring paradigms). Both converge to 0% full-game solve rate, motivating pivot to the ensemble approach.
- **Early March 2026:** Implementation of Ensemble-Itr-1: 4-signal combinatorial ensemble (contextual, lexical, WordNet, entity-based) achieving 16% game accuracy. Mid-submission report completed.
- **Mid March 2026:** Expansion to the full 17-signal ensemble across three signal families (lexical, neural, knowledge-graph). LightGBM LambdaRank integrated. Contrastive fine-tuning with hard-negative mining implemented.
- **Late March 2026:** Data augmentation pipelines completed BabelNet knowledge-graph-based (500 puzzles) and LLM-based (100 puzzles). Augmented training run executed with 1,278 total training games.
- **Early April 2026:** Ablation studies designed and begun. Full evaluation on 100-game test set completed: **33% game accuracy, 50.25% group accuracy**. Final report writing.
- **10th April 2026:** Submitted the code, Report and Presentation.

The GNN exploration (February 2026) surfaced three decisive architectural insights that shaped the rest of the project: the global optimization requirement, the inadequacy of sparse knowledge graphs, and the necessity of heterogeneous signal fusion. The GNN branch was ultimately abandoned in favor of the more tractable ML ensemble approach.

6 Results and Analysis

6.1 Analysis

The progression from Baseline 0 to the Final Ensemble reveals several decisive insights.

Embedding architecture matters enormously. Moving from static GloVe vectors (Baseline 0: 1%) to contextual MPNet embeddings with double centering (Baseline 2: 11%) yields an *11×* improvement on full-game solve rate, confirming that context-sensitive representations are non-negotiable for a task where polysemy is deliberately weaponized.

GNN approaches fundamentally misalign with the problem structure. The GNN collapse to 0% exposes a theoretical mismatch: lateral wordplay (homophones,

Architecture / Model	Games Solved	Groups Solved	Groups “One Away”	Games “One Away”
Baseline 0 (GloVe + K-Means)	1%	10.50% (42/400)	33.25% (133/400)	8%
Baseline 1 (GloVe + SBERT Agglo.)	0%	22.50% (90/400)	19.75% (79/400)	0%
Baseline 2 (MPNet + Double Center.)	11%	24.25% (97/400)	43.25% (173/400)	3%
GNN (Node / Global Opt.)	0%	3.00% (12/400)	19.00% (76/400)	0%
Ensemble-Itr-1 (4 signals)	16%	33.50% (134/400)	43.50% (174/400)	10%
Final Ensemble (Augmented)	33%	50.25% (201/400)	32.00% (160/400)	7%

Table 1: Comparative evaluation of all our architectures on the 100-puzzle held-out test set (4-guess budget).

Model	Guess	Game Accuracy
MPNet(10)	5	11.6%
GPT-3.5 Turbo (10)	5	6.42%
GPT-4 Turbo (10)	5	29.2%
GPT-4 Turbo + CoT (10)	5	38.93%
Ours: Final Ensemble	4	33.00%

Table 2: Comparison with LLM baselines from Todd et al. Note that all Todd et al. figures use a relaxed 5-guess budget; our model achieves 33% under the stricter 4-guess game constraint.

shared prefixes, cultural associations) does not naturally form knowledge-graph edges, making GNNs structurally blind to the most challenging puzzle categories.

Heterogeneous signal fusion is the decisive factor. Ensemble-Itr-1’s jump to 16% with only 4 signals validates the multi-signal hypothesis. The final 17-signal ensemble achieves 33% a further **2.06× improvement** confirming that phonetic, morphological, and grid-aware uniqueness signals each resolve a distinct class of puzzle category.

Data augmentation provides substantial generalization lifting. Training on 1,278 augmented puzzles vs. 678 original gives a measured lift in accuracy without contaminating the evaluation split. The LambdaRank NDCG@1 on validation reaches 0.92, indicating the ranker correctly identifies the top group for over 90% of validation games.

The “One Away” rate as diagnostic. The Baseline 2 and Ensemble-Itr-1 “One Away” rates (43%) indicate the models are semantically sound but fail at final disambiguation. The Final Ensemble’s reduced rate (32%) alongside dramatically higher group accuracy suggests the grid-aware uniqueness matrices and global optimizer successfully resolve these ambiguities.

Comparison with LLM baselines (Table 2). Todd et al.(10) report GPT-4 Turbo achieving 29.2% and GPT-4 Turbo with Chain-of-Thought (CoT) achieving 38.93%, both evaluated under a relaxed 5-guess budget. Our Final Ensemble achieves **33%** under the strictly harder **4-guess budget** surpassing GPT-4 Turbo alone and approaching GPT-4+CoT without any access to a large language model during inference. This result is especially significant because our approach is fully transparent and deterministic, whereas LLM-based methods rely on opaque probabilistic generation. The MPNet

embedding baseline from Todd et al. (11.6%, 5 guesses) closely mirrors our own Baseline 2 (11%, 4 guesses), providing strong cross-study validation of our experimental setup.

6.2 Ablation Study

To quantify the contribution of each signal family and architectural component, we conduct a **progressive signal-inclusion ablation**: starting from a minimal pure-semantic baseline, we incrementally add signal families until reaching the full model. All variants share the same ML ensemble (RF + GBM + LightGBM LambdaRank + LR Fusion), the same 678-game original training split, and the identical sealed 100-game test set.

6.2.1 Ablation Analysis

Semantic embeddings alone are insufficient but essential (V1: 21%, 39.25%). Even with only three neural signals (MPNet, GloVe, WordNet), the pure-semantic baseline already solves 21% of games validating that contextual embeddings capture the primary thematic grouping signal. However, the 39.25% group accuracy confirms that a large fraction of groups share only shallow semantic overlap, requiring richer relational evidence.

Dense knowledge and lexical signals provide a clear but bounded lift (V2: 25%, 41.75%). Adding ConceptNet Numberbatch (knowledge-retrofitted embeddings) and the three lexical signals (character n-gram, phonetic edit distance, morphological affix overlap) yields a +4% game accuracy gain and +2.5% group accuracy. The gain is bounded because these signals still lack access to external world-knowledge databases phonetics helps for homophones, and Numberbatch helps for common-sense pairs, but encyclopedic category memberships are not yet captured.

Neural context signals are the single largest driver (V3: 32%, 50.50%). Adding multi-sense embeddings, pairwise context similarity, and template-context similarity produces the sharpest jump in the ablation: +7% game accuracy and nearly +9% group accuracy. This confirms that *contextual disambiguation* knowing which sense of a word is relevant on a particular board, and whether two words appear together in structured sentences is the most informative signal family for puzzle solving.

External knowledge graphs alone cause a temporary regression (V4: 29%, 48.25%). Counter-

Variant	Active Signal Families	Matrices	Features	Game Acc.	Group Acc.
V1 — Pure Semantic Baseline	MPNet, GloVe, WordNet	3	27-d	21.00%	39.25%
V2 — + Dense KG Embeddings	V1 + Numberbatch, Lexical, Phonetic, Morphological	7	63-d	25.00%	41.75%
V3 — + Neural Context + LR Fusion	V2 + MultiSense, PairContext, TemplateCtx, LR Fusion	10	90-d	32.00%	50.50%
V4 — Full Signals, Original Data	All 17 signals (ext. KGs + Grid-Aware Uniqueness)	17	153-d	29.00%	48.25%
V5 — Full Signals, RF+GBM Only	All 17 signals, LambdaRank disabled	17	153-d	33.00%	50.50%
Final (Augmented + Full Ensemble)	All 17 signals + 600 augmented training puzzles	17	153-d	33.00%	50.25%

Table 3: Progressive ablation on the sealed 100-puzzle test set (4-guess budget).

intuitively, adding the full set of external knowledge-graph signals (Wikidata, ConceptNet, Wikipedia categories, and Grid-Aware Uniqueness matrices) to the full 17-matrix model *without* augmented data slightly hurts performance relative to V3. This is likely due to noise introduced by sparse API responses on a modest 678-game training set the ML classifiers overfit to noisy KG signals that do not generalize, a manifestation of the high-dimensionality/small-data problem that motivates augmentation.

Removing LambdaRank recovers lost accuracy (V5: 33%, 50.50%). When the LightGBM LambdaRank ranker is disabled and only RF + GBM are used with the full 17-signal feature set (V5), game accuracy recovers to 33% matching the augmented final model. This indicates that the LambdaRank ranker, trained on only 678 games with 153 features, is prone to overfitting on the sparse KG signals. The ranker provides strong contributions only when trained on a sufficiently large and diverse corpus.

Data augmentation stabilizes and unlocks the full 17-signal model (Final Augmented: 33%, 50.25%). Expanding the training corpus to 1,278 puzzles (adding 600 augmented games) allows the full ensemble including LambdaRank to train stably on all 17 signals without overfitting. The final model matches V5’s game accuracy while using the complete ensemble and achieving reliable LambdaRank NDCG@1 of 0.92 on validation.

7 Future Work

The multi-signal ensemble framework developed in this project opens four compelling directions for continued research and real-world application.

1. Cross-Domain Word Association Benchmarking. The architecture generalizes beyond NYT *Connections* to any lateral grouping task involving structured semantic constraint satisfaction. For example, in **medical NLP**, the same pipeline could group drug names by shared mechanism of action (e.g., all ACE inhibitors vs. all beta-blockers) despite surface-level name dissimilarity.

2. LLM-Augmented Reasoning Integration. Our current ensemble is being extended with a lightweight language model (LLM) reasoning layer that takes the top-scored candidate partition from the ensemble and uses chain-of-thought prompting to verify or refine each group assignment. For example, given the candidate group {Mercury, Venus, Earth, Pluto}, a small reasoning model can confirm “all are planets/former planets

in the Solar System” or flag the distractor. This hybrid approach ensemble scoring for efficiency, LLM reasoning for precision targets the adversarial top-difficulty categories that pure signal fusion cannot resolve.

3. Active Learning from Human Gameplay Data.

Each time a human player makes an incorrect guess in the NYT *Connections* game, it provides a high-quality hard negative: a 4-word group the human found plausible but is factually incorrect (e.g., guessing {Bass, Trout, Drum, Sole} when *Drum* and *Sole* are actually in the “musical instruments” and “shoe parts” categories). Feeding such gameplay signals back into the LambdaRank and GBM training loop offers a principled online learning mechanism that directly models human cognitive biases and distractor susceptibility.

4. Adversarial Puzzle Generation for Stress-Testing. A controllable puzzle generator that injects precisely calibrated semantic distractors e.g., inserting one semantically adjacent “trap” word into each group would create a systematic benchmark for measuring model robustness under varying distractor pressure. Such a generator could also be used to automatically surface the weakest signal channels in the ensemble (e.g., phonetic similarity performs well on homophones like {Sole, Soul, Seoul, Sell} but fails on purely cultural connection groups), guiding targeted architectural improvements.

8 Conclusion

This work systematically investigated the challenge of solving the NYT *Connections* puzzle through explicit semantic modeling and combinatorial optimization. Starting from simple GloVe clustering baselines, we progressed through GNN exploration to a final 17-signal ensemble architecture trained on an augmented corpus of 1,278 puzzles.

What We Achieved:

- A **33× improvement** in full-game accuracy over the GloVe baseline (from 1% to 33%) and a **3× improvement** over MPNet baseline (11%).
- A 50.25% group accuracy solving, on average, exactly 2 of the 4 groups per game on unseen puzzles.
- A principled multi-modal feature engineering framework (153 features across 17 heterogeneous matrices) that provably benefits from signal diversity, validated by LR matrix-weight coefficients

showing positive contributions from 14 of 17 signals.

- A rigorous data augmentation pipeline generating 600 structurally valid novel puzzles, enabling $2\times$ training data expansion without contaminating the evaluation split.
- A globally optimal solver that prevents the cascade failures of greedy clustering, validated by the drop in “One Away” rate from 43% to 32%.

What We Could Not Achieve and Why:

- **Human-level performance** remains out of reach. The hardest NYT *Connections* puzzles (purple/black difficulty tiers) involve pure cultural knowledge references, wordplay homonyms, and structural patterns with zero distributional signal categories where even our grid-aware uniqueness matrices provide negligible signal, as Wikidata and Wikipedia often lack entries for colloquial or highly specific puzzle categories.
- **GNN approaches** were decisively abandoned. Despite significant engineering effort (loss balancing, skip connections, symmetric feature extraction), the fundamental sparsity of knowledge-graph edges for lateral wordplay tasks renders GNNs structurally incompatible with the problem. This is a theoretical constraint, not a practical limitation.
- **The LLM augmentation quality ceiling:** The 100 LLM-generated puzzles, while structurally valid, have lower lateral challenge than human-authored puzzles they lack the deliberate adversarial “red herring” design that makes the original puzzles hard. Future augmentation methods must explicitly generate controlled semantic distractors.
- **Dataset size remains a binding constraint.** Even after augmentation, our effective training corpus of 1,278 puzzles (5,112 positive group samples) is modest for tree-based ensembles such as Random Forest, which are inherently data-hungry each of the 200 decision trees must generalize over 153 features with limited examples per leaf. Due to computational resource constraints (each training run requires ~ 2 hours of API-heavy feature extraction), we could not scale augmentation substantially further. A larger corpus of diverse, human-quality puzzles would likely push group accuracy well beyond the 50% ceiling observed here.

Taken together, these results establish that heterogeneous linguistic signal fusion under global combinatorial optimization is the correct architectural choice for word association puzzles not deep learning alone, not GNNs alone, and not LLMs alone.

References

- [1] Chen, Z., et al. (2024). Towards explaining the power of constant-depth graph neural networks for structured linear programming. In *ICLR 2024*.
- [2] Lopez, A. Y. L., McDonald, T., & Emami, A. (2024). NYT-Connections: A deceptively simple text classification task that stumps system-1 thinkers. *arXiv:2412.01621*.
- [3] Merino, T., et al. (2024). Making new connections: LLMs as puzzle generators for the New York Times’ Connections word game. *arXiv:2407.11240*.
- [4] Mikolov, T., et al. (2013). Efficient estimation of word representations in vector space. *arXiv:1301.3781*.
- [5] Miller, G. A. (1995). WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41.
- [6] Samadarshi, P., et al. (2024). Connecting the dots: Evaluating abstract reasoning capabilities of LLMs using the New York Times Connections word game. *arXiv:2406.11012*.
- [7] Schlichtkrull, M., et al. (2018). Modeling relational data with graph convolutional networks. In *European Semantic Web Conference*.
- [8] Song, K., et al. (2020). MPNet: Masked and permuted pre-training for language understanding. *arXiv:2004.09297*.
- [9] Speer, R., Chin, J., & Havasi, C. (2017). ConceptNet 5.5: An open multilingual graph of general knowledge. *arXiv:1612.03975*.
- [10] Todd, G., et al. (2024). Missed connections: Lateral thinking puzzles for large language models. *arXiv:2404.11730*.
- [11] Zhang, S., et al. (2023). Optimizing over trained GNNs via symmetry breaking. *arXiv:2305.09420*.

A Project Links and Resources

- **GitHub Repository:** https://github.com/Anurag-Kacholiya/NYT-Connection_Solver
- **Hugging Face Models:** <https://huggingface.co/Rimuru2303/NYT-Connections/tree/main>
- **Presentation Slides:** https://drive.google.com/drive/u/0/folders/11t9isZ1m1p6MneZuCg_7ssG2bap4rDyY
- **Demonstration Video:** https://drive.google.com/drive/u/0/folders/1qbecz0hAb_vRZP61-aaVL9GnUv6db60X
- **Step-by-Step Code Workflow Walkthrough:** https://srushtipekamwar.github.io/INLP_Project_Final_Model/project_report.html